Deep Learning for Limit Order Books

Neural and Spatial Models for Short-Term Limit Order Book Dynamics

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Abstract

This report studies neural network methods for predicting short-term limit order book movements. Following the formulation of Sirignano [7], we model the joint distribution of future best ask and best bid movements, rather than only predicting the direction of the mid-price. Due to data and computational constraints, we perform a reduced replication using the publicly available WSELOB-2017 dataset. From the raw order book event data, we reconstruct fixed-frequency snapshots and define a probabilistic multiclass prediction problem over clipped joint quote movements.

We compare several models of increasing complexity: a naive empirical baseline, multinomial logistic regression, a standard feedforward neural network, a reduced spatial neural network and a fuller local spatial neural network. The models are evaluated using negative log-likelihood and movement-conditional negative log-likelihood, since ordinary accuracy is misleading in this highly imbalanced setting.

The results show that the current limit order book state contains useful predictive information. Logistic regression improves upon the empirical baseline for all stocks, and neural network models improve further. The standard neural network performs well in terms of overall likelihood, while the reduced spatial neural network is especially competitive on observations where the best ask or best bid actually moves. The full local spatial neural network provides additional movement-specific improvements for some stocks, but is less stable overall. Compared with the original paper, our findings are therefore more nuanced: spatial structure is useful for modelling nonzero quote movements, but does not uniformly dominate a standard neural network on the WSELOB-2017 data.

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1
Introduction

Modern financial markets are largely organized through electronic limit order books. At any point in time, the limit order book contains the outstanding buy and sell orders submitted by market participants, ordered by price priority. The best bid and best ask prices determine the immediately available trading prices, while the deeper levels of the book describe the available liquidity away from the current market price. As a result, the limit order book contains a detailed high-frequency description of supply and demand. Understanding whether this information can be used to predict short-term price movements is relevant for several areas of financial computing, including market making, execution algorithms, short-term risk management and market microstructure research.

Predicting limit order book dynamics is challenging for several reasons. The data are high-dimensional, noisy and highly imbalanced: over short time horizons, the best bid and best ask often do not move at all. Moreover, the relevant information is not only contained in the best quote, but also in the shape of the book across multiple price levels. This makes limit order book prediction a natural setting for machine learning methods, and in particular for neural networks that can learn nonlinear relationships from high-dimensional inputs.

A key contribution in this direction is the work of Sirignano [7], who studies deep learning methods for predicting future limit order book states. Instead of only predicting the direction of the mid-price, the paper models the joint distribution of future best ask and best bid movements. This is a more complete prediction problem: the best ask and best bid do not necessarily move together, and separate movements of the two sides of the book determine changes in the spread. The paper compares simple baselines, standard neural networks and a spatial neural network architecture that is designed to exploit the local structure of price levels in the limit order book.

The goal of this project is to perform a reduced replication and extension of this idea. A full replication of the original study is not feasible within the scope of this course, since the original work relies on a very large proprietary high-frequency dataset and substantial computational resources. Instead, we focus on reproducing the main modelling idea on a smaller scale. We study whether the current state of a reconstructed limit order book can be used to estimate the conditional distribution of short-term best ask and best bid movements.

Let A_t and B_t denote the best ask and best bid prices at time t, respectively, and let X_t denote the observed state of the limit order book. For a fixed prediction horizon Δt, we define the target variable

$$Y_t = (\Delta A_t, \Delta B_t),$$

where

$$\Delta A_t = \frac{A_{t+\Delta t} - A_t}{\text{tick size}}, \qquad \Delta B_t = \frac{B_{t+\Delta t} - B_t}{\text{tick size}}.$$

Thus, the statistical learning problem is to estimate

$$\mathbb{P}(Y_t = y \mid X_t),$$

for possible joint movements y. In our implementation, the output space is made finite by clipping both components of Y_t to the range $\{-K, \ldots, K\}$. This turns the problem into a multiclass probabilistic prediction task with $(2K+1)^2$ possible joint movement classes.

We compare several models of increasing complexity. First, we use an empirical baseline that ignores the current order book state and predicts only the unconditional distribution of future movements. Second, we use multinomial logistic regression as a linear conditional model. Third, we train a standard feedforward neural network to test whether nonlinear functions of the order book state improve predictive performance. Finally, we consider two spatial neural network variants: a reduced spatial model based on a factorized hazard-style representation, and a fuller local spatial model that uses local windows around candidate price levels.

The main focus of the empirical analysis is not ordinary classification accuracy. Since many observations correspond to no movement of the best bid or ask, a model can obtain high accuracy by predicting the no-movement class almost always. We therefore evaluate the models primarily using negative log-likelihood, which measures the quality of the full predicted probability distribution. In addition, we report movement-conditional negative log-likelihood, which evaluates model performance only on observations where the best bid or best ask actually changes.

The remainder of this report is structured as follows. Section 2 introduces the limit order book setting and formalizes the prediction problem. Section 3 describes the baseline models, the standard neural network and the spatial neural network variants. Section 4 discusses the dataset, preprocessing pipeline, training procedure and evaluation metrics. Section 6 presents and interprets the empirical results. Finally, Section 7 summarizes the main findings and discusses possible directions for future work.

2
Background and Problem Setup

This chapter introduces the basic concepts needed to formulate the prediction problem studied in this project. We first give a short overview of how limit order books work and define the main quantities used throughout the report, such as the best bid, best ask, spread and order book depth. We then formalize the prediction task as a probabilistic learning problem. We finish this chapter of by looking at some related works.

2.1. Limit Order Books

A limit order book is the mechanism through which many electronic exchanges organize trading. Market participants can submit buy and sell orders for a given asset. A buy order specifies a maximum price at which the trader is willing to buy, while a sell order specifies a minimum price at which the trader is willing to sell. Orders that are not immediately executed remain in the book as standing limit orders.

The buy side of the book is called the bid side, and the sell side is called the ask side. The highest available buy price is called the best bid price, denoted by B_t, and the lowest available sell price is called the best ask price, denoted by A_t. Since a trade can only occur when a buyer is willing to pay at least as much as a seller is willing to accept, the best bid is normally below the best ask:

$$B_t < A_t.$$

The difference between these two prices is the bid-ask spread,

$$S_t = A_t - B_t.$$

The spread is an important measure of trading costs and liquidity. A small spread usually indicates that buyers and sellers are close in price, while a large spread indicates a less liquid or more uncertain market.

A common summary price is the mid-price,

$$M_t = \frac{A_t + B_t}{2}.$$

Many prediction problems in the limit order book literature focus on the future direction of the mid-price. However, the mid-price hides some of the structure of the book. For example, the best bid and best ask can move separately, causing the spread to widen or narrow even if the mid-price changes only slightly. For this reason, the original paper models the joint movement of the best ask and best bid, rather than only the movement of the mid-price.

The order book contains information not only at the best bid and best ask, but also at deeper price levels. On the ask side, the first level corresponds to the best ask price A_t, the second level corresponds to the next lowest ask price, and so on. Similarly, on the bid side, the first level corresponds to the best bid price B_t, the second level corresponds to the next highest bid price below the best bid, and so on. If L levels are observed on each side of the book, we may write the book state schematically as

$$X_t = \left(p_t^{a,1}, v_t^{a,1}, \ldots, p_t^{a,L}, v_t^{a,L}, p_t^{b,1}, v_t^{b,1}, \ldots, p_t^{b,L}, v_t^{b,L}\right),$$

where $p_t^{a,i}$ and $v_t^{a,i}$ denote the price and available volume at ask level i, and $p_t^{b,i}$ and $v_t^{b,i}$ denote the price and available volume at bid level i. The volumes at different price levels describe the available liquidity in the book. If there is a large amount of volume close to the best ask, for instance, then upward price movements may be harder because incoming buy orders need to consume this liquidity before the best ask can increase. Conversely, if the ask side is thin near the best price, a relatively small amount of buying pressure may move the best ask upward. Similar reasoning applies to the bid side. This motivates using the full shape of the order book as input for prediction.

A useful derived quantity is order book imbalance. At the best level, this can be written as

$$I_t = \frac{v_t^{b,1} - v_t^{a,1}}{v_t^{b,1} + v_t^{a,1}}.$$

If I_t is positive, there is more volume available at the best bid than at the best ask. If I_t is negative, the opposite is true. Imbalance-type features are often used because they summarize the relative pressure between the buy and sell sides of the book. In our implementation, similar imbalance features are computed level by level.

The state of the limit order book changes whenever market participants submit, cancel, modify or execute orders. These changes occur at very high frequency and create a dynamic picture of supply and demand. The central question in this project is whether the current book state X_t, including volumes and relative price levels, contains useful information about the future movement of the best ask and best bid prices.

2.2. Prediction Problem

We now formalize the prediction problem considered in this project. Following the formulation in [7], the objective is to estimate the joint distribution of future best ask and best bid movements conditional on the current state of the limit order book. Let A_t and B_t denote the best ask and best bid prices at time t, and let X_t denote the observed limit order book state. In our implementation, X_t contains information from the first L levels on both sides of the book, including transformed volumes, relative price levels, spread and imbalance-type features. For a fixed prediction horizon Δt, we define the target variable as

$$Y_t = (\Delta A_t, \Delta B_t),$$

where the two components are measured in tick units:

$$\Delta A_t = \frac{A_{t+\Delta t} - A_t}{\text{tick size}}, \qquad \Delta B_t = \frac{B_{t+\Delta t} - B_t}{\text{tick size}}.$$

Thus, Y_t describes the joint movement of the best ask and best bid over the prediction horizon.

In the original spatial neural network formulation of Sirignano [7], the objective is not restricted to a finite set of movement classes. Instead, the spatial neural network is designed to model a probability distribution on the full spatial domain, so that movements are not artificially truncated to a bounded region. This is an important conceptual difference from a standard multiclass neural network, which can only assign probabilities to a fixed finite set of classes. In principle, such a full-space formulation is more faithful to the original prediction problem, since large future movements can still be represented explicitly rather than being absorbed into boundary classes. In practice, however, implementing and training this full distributional model was not computationally feasible within the scope of this project. It would require evaluating and normalizing probabilities over a much larger, effectively unbounded output space, while also training several models across multiple stocks. We therefore use a finite approximation. This turns the problem into a tractable probabilistic multiclass prediction task, while preserving the main modelling objective of estimating the conditional joint distribution of future best ask and best bid movements.

To obtain a finite output space, both price movements are clipped to a fixed range. For a chosen integer K, we define

$$\mathcal{Y} = \{-K, \dots, -1, 0, 1, \dots, K\}^2.$$

If the absolute movement of either component exceeds K ticks, it is assigned to the corresponding boundary class. In the experiments we use $K = 10$, resulting in

$$|\mathcal{Y}| = (2K + 1)^2 = 441$$

possible joint movement classes. The statistical learning problem is therefore to estimate

$$\mathbb{P}(Y_t = y \mid X_t), \qquad y \in \mathcal{Y}.$$

This is a probabilistic multiclass prediction problem: the goal is not only to predict the most likely future movement, but to estimate the full conditional distribution over all possible clipped joint movements. For implementation purposes, each pair $y = (\Delta A_t, \Delta B_t) \in \mathcal{Y}$ is mapped to a single class label. With K denoting the clipping level, the class index is defined as

$$c(y) = (\Delta A_t + K)(2K + 1) + (\Delta B_t + K).$$

The models are then trained to assign high probability to the observed class label. Throughout the project, we compare different estimators of this conditional distribution, ranging from simple empirical and linear baselines to neural-network-based models.

2.3. Related Work

Limit order book prediction has been studied using a range of statistical and machine learning methods. A classical direction is based on hand-crafted market microstructure features, such as order book imbalance and order flow imbalance. These features summarize the relative pressure between the bid and ask side of the book and have been shown to contain short-term predictive information about price changes [1, 2]. This line of work is relevant for our project because several of our input features, such as level-wise imbalances, are motivated by the same idea that the shape of the book contains information about future quote movements.

Deep learning methods have become increasingly common for limit order book prediction. Tsantekidis et al. [9] use convolutional neural networks to predict short-term price movements from limit order book data. Zhang, Zohren and Roberts [10] propose DeepLOB, an architecture combining convolutional layers and recurrent layers to capture both spatial structure across price levels and temporal dependence in the order book. Tran et al. [8] introduce a temporal attention-augmented bilinear network, showing that attention mechanisms can help identify informative time points in financial time-series data. These papers are related to our work because they all use neural networks to extract predictive information from high-dimensional order book states.

Another closely related contribution is the work of Kolm, Turiel and Westray [4]. They study deep learning models for high-frequency return forecasting at multiple horizons and show that order-flow based inputs can be more effective than raw order book states. Their work is particularly relevant because it highlights that the representation of the order book can matter as much as the neural network architecture itself. Their use of stock-specific horizons based on average price changes is also relevant to our findings, since the WSELOB stocks differ substantially in activity level and the fixed horizon used in this project may not be equally suitable for all assets.

The paper most directly related to this project is Sirignano's work on deep learning for limit order books [7]. The key idea we take from this paper is to model the joint distribution of future best ask and best bid movements, and to investigate whether spatial structure in the order book can improve probabilistic prediction. Our project follows this formulation, but studies a reduced replication with simplified architectures and a different data source.

3
Spatial Structure and Model Methodology

This chapter introduces the models used in the empirical comparison. We first discuss the local spatial structure of limit order books, which motivates the spatial architectures used later. We then present two baseline models, followed by a standard neural network and two spatial neural network variants inspired by [7].

3.1. Local Spatial Structure in Limit Order Books

A limit order book has a natural spatial structure. Price levels are ordered relative to the current best bid and best ask, and nearby levels are more closely related than distant levels. For example, a movement of the best ask by one tick is more closely related to a movement of two ticks than to a movement of ten ticks. This ordering is important because larger price movements must pass through intermediate price levels. A standard multiclass neural network does not explicitly use this structure. It treats each possible joint movement $(\Delta A_t, \Delta B_t)$ as a separate output class, even though neighbouring movement classes are naturally related.

The spatial neural network proposed by Sirignano [7] is designed to address this by exploiting the local structure of price levels in the book. The main intuition is that the probability of a price moving through a certain level should depend mainly on the liquidity close to that level. If there is a large amount of volume near the best ask, an upward ask movement is less likely, since this liquidity must first be consumed. Conversely, if the book is thin near the best price, a small amount of order flow may already move the quote. This motivates a level-by-level, or hazard-style, view of price movements. Instead of directly assigning unrelated probabilities to all movement sizes, the model can represent a movement as a sequence of continuation decisions: the price either stops at a certain level or continues through it. In this project, we use this idea in two forms: a reduced spatial neural network with a simpler factorized hazard structure, and a fuller local spatial neural network that uses local liquidity windows around candidate price levels.

The spatial structure discussed in this section can be understood directly from the mechanics of the limit order book. In order for the best ask or best bid to move by several ticks, liquidity at the intermediate price levels must first be consumed. A one-tick upward move in the best ask only requires the first ask level to be removed, whereas a two-tick move requires both the first and second ask levels to be consumed, and so on. This implies that the probability of a larger quote movement depends not only on the best quote itself, but also on the liquidity available deeper in the book. Figure 3.1 illustrates this intuition schematically and motivates why neighboring movement classes should be related.

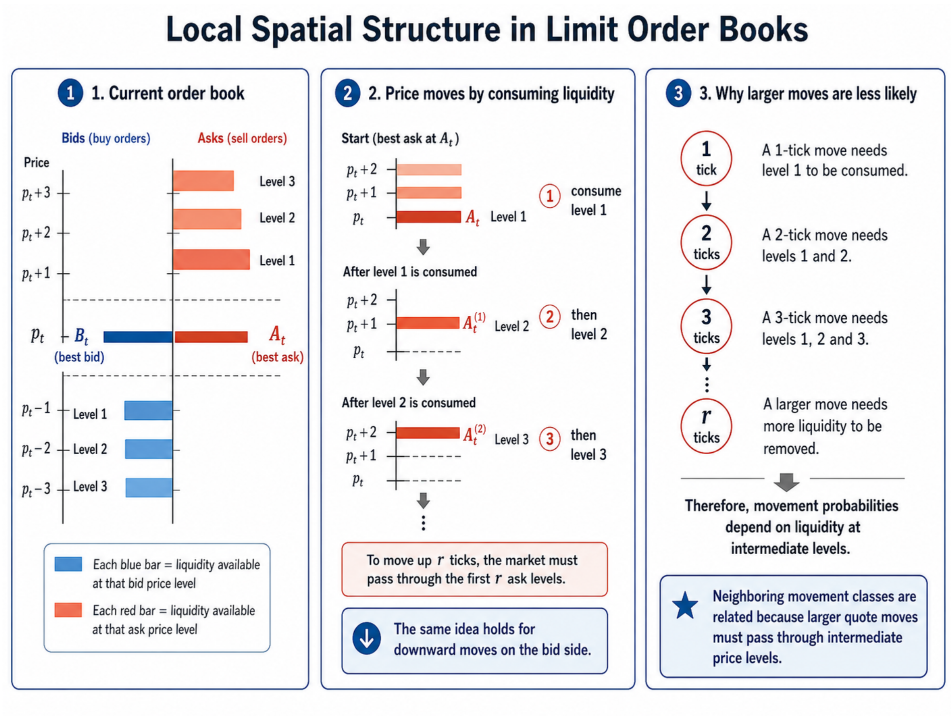

Figure 3.1: Schematic illustration of local spatial structure in a limit order book. The left panel shows the current order book, the middle panel illustrates how an upward price move requires successive consumption of liquidity at intermediate ask levels, and the right panel explains why larger quote movements are less likely. Since an r-tick movement must pass through the first r price levels, movement probabilities depend on liquidity not only at the best quote, but also deeper in the book.

3.2. Baseline Models

Before considering neural network models, we first introduce two baseline models. These baselines provide reference points for evaluating whether more complex models actually extract useful information from the limit order book. The empirical model ignores the current book state and only captures the unconditional distribution of future movements. Logistic regression then serves as a simple conditional model, using the order book features but restricting the relationship between features and class probabilities to be linear.

3.2.1. Naive Empirical Model

The first baseline is a naive empirical model. This model ignores the current limit order book state X_t and predicts future movements only from their empirical frequencies in the training set. It therefore estimates the unconditional distribution of the target variable Y_t.

Let $\mathcal{D}_{\text{train}} = \{Y_i\}_{i=1}^{N}$ denote the training labels. For each possible joint movement $y \in \mathcal{Y}$, the empirical model assigns probability

$$\hat{p}_{\text{emp}}(Y = y) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{Y_i = y\}.$$

This probability is then used for every test observation, regardless of the corresponding order book state:

$$\hat{p}_{\text{emp}}(Y_t = y \mid X_t) = \hat{p}_{\text{emp}}(Y = y).$$

Although this model is very simple, it is an important reference point. If a conditional model does not improve upon the empirical baseline, then it has not extracted useful predictive information from the limit order book features. The empirical model also reflects the strong class imbalance in the data, since the no-movement class $(0, 0)$ often has the highest probability.

3.2.2. Logistic Regression

The second baseline is multinomial logistic regression. Unlike the empirical model, logistic regression uses the current order book state X_t to predict the distribution of future movements. It therefore provides a simple conditional baseline against which the neural network models can be compared. Since the clipped target space $\mathcal{Y}$ is finite, each joint movement $y \in \mathcal{Y}$ is represented by a class label $c \in \{1, \dots, C\}$, where $C = (2K+1)^2$. Logistic regression assigns a linear score to each class,

$$z_c(X_t) = \theta_c^\top X_t + b_c,$$

where θ_c and b_c are trainable parameters. These scores are transformed into class probabilities using the softmax function:

$$p_\theta(Y_t = c \mid X_t) = \frac{\exp(z_c(X_t))}{\sum_{j=1}^{C} \exp(z_j(X_t))}.$$

The model is trained by minimizing the negative log-likelihood on the training set,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \log p_\theta(Y_i \mid X_i).$$

This is the same probabilistic objective used for the neural network models later in the report.

Although the logistic regression model is linear in the feature vector X_t, it is not applied only to raw order book volumes. Following the role of logistic regression in Sirignano [7], we use it as a stronger baseline by including order book features such as relative price levels, spread and imbalance-type features. In particular, imbalance features are nonlinear functions of bid and ask volumes, for example ratios comparing liquidity on the two sides of the book. Thus, logistic regression should be interpreted as a linear classifier on a nonlinear feature representation of the order book. If the neural network models improve upon this baseline, this suggests that they capture additional nonlinear interactions beyond those already included through the constructed features.

3.3. Neural Network Models

The baseline models provide useful reference points, but they are limited in the relationships they can represent. Logistic regression uses the current order book state, but only through a linear function of the input features. Neural networks allow us to replace this linear dependence by a nonlinear function learned from data. In this section, we first introduce a standard feedforward neural network. We then use this model as a stepping stone toward the spatial neural network variants.

3.3.1. Standard Neural Network Model

The standard neural network model estimates the conditional distribution of the clipped joint movement Y_t given the current limit order book state X_t. Since the output space $\mathcal{Y}$ contains $C = (2K+1)^2$ possible classes, the network produces one score, or logit, for each possible joint movement. Let $X_t \in \mathbb{R}^p$ denote the input feature vector. A feedforward neural network with M hidden layers computes

$$h_0 = X_t,$$

and

$$h_m = \phi(W_m h_{m-1} + b_m), \qquad m = 1, \dots, M,$$

where W_m and b_m are trainable parameters. In our implementation, the activation function is the rectified linear unit,

$$\phi(z) = \max(z, 0),$$

applied componentwise. The final layer maps the last hidden representation to C logits:

$$z_\theta(X_t) = W_{M+1} h_M + b_{M+1}.$$

These logits are converted into class probabilities using the softmax function:

$$p_\theta(Y_t = c \mid X_t) = \frac{\exp(z_{\theta,c}(X_t))}{\sum_{j=1}^{C} \exp(z_{\theta,j}(X_t))}, \qquad c = 1, \dots, C.$$

Thus, the network outputs a full probability distribution over all possible clipped joint movements, rather than only a single predicted class.

The model is trained by minimizing the negative log-likelihood on the training set,

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log p_\theta(Y_i \mid X_i).$$

This is the same probabilistic objective used for logistic regression, but the neural network can capture nonlinear interactions between volumes, price levels, spread and imbalance features.

In the experiments, this standard neural network is implemented as a modest two-layer multilayer perceptron. Both hidden layers have width 128 and are followed by a ReLU activation and dropout with rate 0.2. The final linear layer produces $C = 441$ logits, corresponding to the clipped joint movement classes. This architecture was chosen to provide a nonlinear but still relatively straightforward benchmark.

The standard neural network is an important benchmark because it tests whether nonlinear modelling improves prediction. However, it still treats the output classes as unrelated categories. It does not explicitly use the fact that neighbouring price movements are spatially related. This motivates the spatial neural network models introduced next.

3.3.2. Reduced Spatial Neural Network

The standard neural network in the previous section is a useful nonlinear benchmark, but it still treats the $C = (2K+1)^2$ joint movement classes as unrelated output categories. This ignores the spatial intuition from Section 3.1: a movement of two ticks is naturally related to a movement of one tick, since larger quote movements must pass through intermediate price levels. As a first step toward the spatial modelling idea of Sirignano [7], we therefore implement a reduced spatial neural network.

The model keeps the clipped prediction problem from Section 2.2, so the target space remains

$$\mathcal{Y} = \{-K, \ldots, K\}^2.$$

This is an important simplification compared with the original spatial neural network. In Sirignano's formulation, the spatial architecture is designed to model a more general distribution over future price locations and to use local maps around candidate future prices. In our reduced implementation, we do not yet use such local windows. Instead, we introduce spatial structure only through the output representation: movement sizes are modelled as ordered, level-by-level events rather than as unrelated classes. This makes the model computationally manageable while preserving the main idea that neighbouring movement sizes should be related.

We factorize the joint distribution into an ask component and a bid component,

$$p_\theta(\Delta A_t, \Delta B_t \mid X_t) = p_\theta(\Delta A_t \mid X_t) p_\theta(\Delta B_t \mid \Delta A_t, X_t).$$

The ask movement is therefore predicted first, and the bid movement is predicted conditionally on both the current order book state and the realized ask movement. This factorization allows the model to represent dependence between the two sides of the book without directly estimating an unstructured probability for all 441 joint classes.

For each one-dimensional movement, the model separates direction and magnitude. First, a neural network predicts whether the movement is negative, zero or positive. Conditional on a nonzero direction, the magnitude is represented by continuation probabilities. For example, for a positive ask movement, let $q_s(X_t)$ denote the probability of continuing beyond level s, conditional on having reached that level. Then, for $1 \le r < K$,

$$p_\theta(\Delta A_t = r \mid X_t) = p_\theta(\Delta A_t > 0 \mid X_t) \left(\prod_{s=1}^{r-1} q_s(X_t)\right) (1 - q_r(X_t)).$$

The boundary class K collects all movements of at least K ticks and receives the remaining continuation probability,

$$p_\theta(\Delta A_t = K \mid X_t) = p_\theta(\Delta A_t > 0 \mid X_t) \prod_{s=1}^{K-1} q_s(X_t).$$

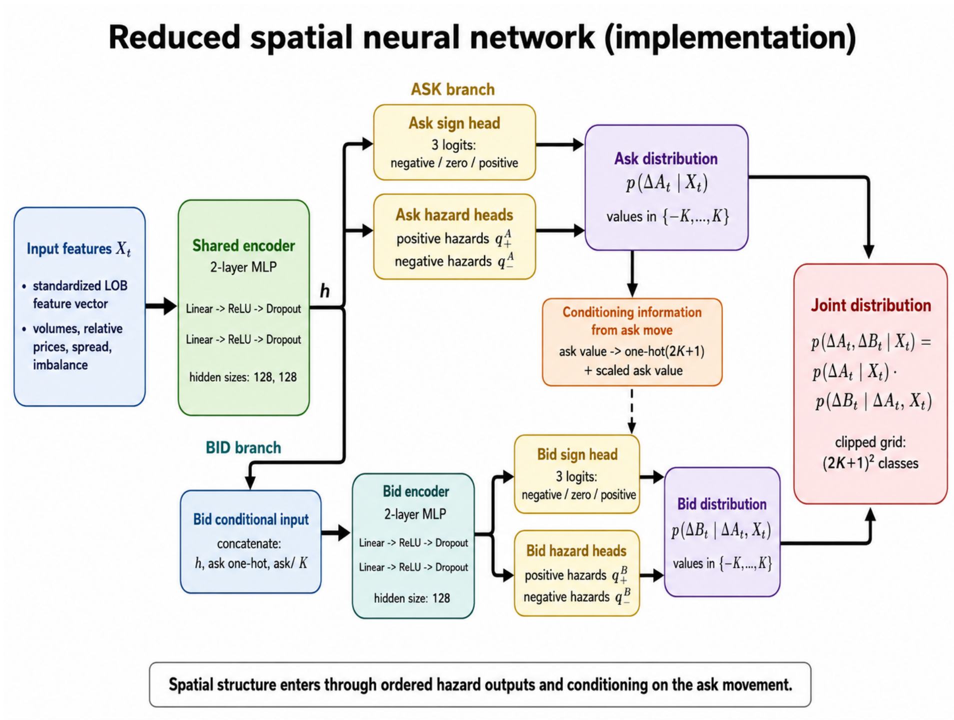

Figure 3.2: Schematic overview of the reduced spatial neural network implementation. The order book state is first encoded into a shared representation. The ask branch predicts $p_\theta(\Delta A_t \mid X_t)$ using sign and hazard outputs. The bid branch then conditions on both the shared book representation and the ask movement to predict $p_\theta(\Delta B_t \mid \Delta A_t, X_t)$. The joint distribution over the clipped grid is obtained by multiplying these two components.

Analogous expressions are used for negative ask movements and for the bid side. In this way, the model encodes the idea that an r-tick movement must pass through the previous $r-1$ levels before stopping at level r.

Figure 3.2 summarizes the architecture of the reduced spatial neural network used in our implementation. The standardized order book feature vector X_t is first passed through a shared encoder, producing a global representation h_t of the current book state. From this representation, the ask branch predicts the distribution of the ask movement by combining a sign head, which assigns probabilities to negative, zero and positive movement, with hazard heads that model the continuation probabilities for positive and negative nonzero movements. The predicted ask movement is then used as conditioning information for the bid branch. Concretely, the bid network receives the shared representation h_t together with an encoding of the ask movement, consisting of a one-hot representation and a scaled ask value. It then predicts the bid sign and bid continuation probabilities in the same hazard-style form. The final joint distribution is obtained by the factorization

$$p_\theta(\Delta A_t, \Delta B_t \mid X_t) = p_\theta(\Delta A_t \mid X_t) p_\theta(\Delta B_t \mid \Delta A_t, X_t).$$

Thus, the model differs from the standard neural network in two ways: it introduces an ordered representation of one-dimensional movement sizes through hazard outputs, and it explicitly conditions the bid movement on the ask movement. However, the continuation probabilities are still computed from a global book representation rather than from local liquidity windows, which is why we refer to this model as a reduced spatial neural network.

3.3.3. Full Local Spatial Neural Network

The reduced spatial neural network in the previous section introduces an ordered hazard-style output representation, but the continuation probabilities are still computed from a global representation of the order book. This means that the model knows that movement sizes are ordered, but it does not yet explicitly inspect the liquidity near the candidate price level being crossed. The full local spatial neural network is introduced to move one step closer to the spatial architecture of Sirignano [7]. The main change is that continuation probabilities are now computed from local liquidity windows around candidate tick levels.

This model keeps the same clipped prediction problem from Section 2.2 and the same joint factorization as the reduced spatial network:

$$p_\theta(\Delta A_t, \Delta B_t \mid X_t) = p_\theta(\Delta A_t \mid X_t) p_\theta(\Delta B_t \mid \Delta A_t, X_t).$$

The ask movement is predicted first, and the bid movement is then predicted conditionally on the ask movement. The difference is not in the factorization itself, but in how the continuation probabilities are computed.

For the full local spatial model, the order book is represented on a dense tick grid. On the ask side, the available volume is stored by tick offset from the current best ask, and on the bid side by tick offset from the current best bid. Missing price levels are assigned zero volume. This dense representation makes it possible to define a local window around each candidate movement level. If s denotes a candidate tick offset, then the model extracts local features of the form

$$m(X_t, s),$$

where $m(X_t, s)$ contains ask-side and bid-side liquidity near level s, together with auxiliary information such as the spread, the normalized candidate offset and a neural representation of the global book state.

The neural network implementation has three main components. First, a global encoder maps the first levels of the dense ask and bid grids, together with the spread, to a global representation of the current book state. In our implementation this encoder is a feedforward network with hidden dimensions 128 and 128. Second, as in the reduced spatial model, separate sign heads predict whether the ask or bid movement is negative, zero or positive. Third, the continuation probabilities are computed by local neural networks applied to the windows $m(X_t, s)$. The same local continuation network is shared across all candidate levels $s = 1, \dots, K-1$. This weight sharing encodes the idea that similar local liquidity patterns should have similar effects at different locations in the book.

For example, for a positive ask movement, the model computes continuation logits for each candidate level $s = 1, \dots, K-1$ using a local network applied to the corresponding window:

$$q_s^A(X_t) = \sigma\big(g_{\theta,+}^A(m(X_t, s))\big)\,.$$

These continuation probabilities are then converted into probabilities over movement sizes using the same hazard-style construction as in Section 3.3.2. Analogous local networks are used for negative ask movements and for positive and negative bid movements. For the bid side, the global representation is first combined with an encoding of the ask movement, so that the bid distribution remains conditional on ΔA_t.

This model is closer to the original spatial neural network than the reduced spatial model because it explicitly evaluates liquidity near the candidate price level. It therefore captures the central spatial intuition from Section 3.1: the probability of moving through a level should depend mainly on the liquidity close to that level. However, it is still not a full reproduction of Sirignano's architecture. The original paper formulates a more general spatial neural network for distributions over future price locations, whereas our implementation is adapted to the finite clipped grid $\{-K, \dots, K\}^2$, the reconstructed WSELOB data and a fixed 30-second horizon. In particular, our local map is implemented as fixed windows on a dense tick grid, rather than as the full general local map used in the original formulation.

Figure 3.3 summarizes the resulting architecture. The figure emphasizes the main difference with the reduced spatial model: the sign prediction still uses a global representation of the book, but the continuation probabilities are now computed from local liquidity windows around candidate tick levels. Thus, the model keeps the same factorization and hazard-style construction as before, while replacing global continuation heads by local continuation networks applied across the dense tick grid.

The full local spatial model is more expressive than the reduced spatial model, but it is also more computationally expensive and more sensitive to class imbalance. For this reason, we retain two checkpoints

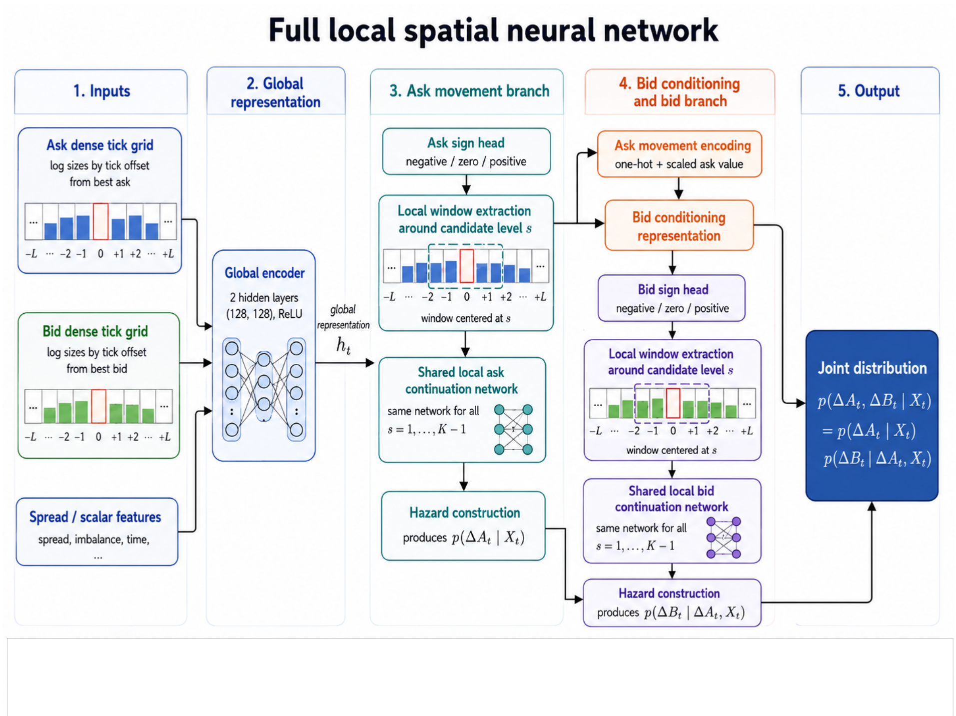

Figure 3.3: Schematic overview of the full local spatial neural network. The order book is represented on dense ask and bid tick grids. A global encoder summarizes the current book state, while local windows around candidate tick levels are used to compute continuation probabilities. The ask branch produces $p_\theta(\Delta A_t \mid X_t)$, and the bid branch conditions on the ask movement to produce $p_\theta(\Delta B_t \mid \Delta A_t, X_t)$. The final output is the joint distribution over clipped ask and bid movements.

during training. The first is selected by the best validation negative log-likelihood over all observations, and the second by the best validation negative log-likelihood restricted to observations with nonzero movement. This allows us to distinguish between overall calibration and performance on the more difficult movement events.

4

Experimental Setup

This chapter describes how the empirical experiments are constructed. We first discuss the choice of dataset and the structure of WSELOB-2017. We then explain how labels and features are constructed. Finally, we describe the training procedure and the evaluation metrics used to compare the models.

4.1. Dataset and Preprocessing

4.1.1. Dataset Availability and Replication Scope

A major practical challenge in limit order book research is data availability. High-quality order book data are expensive, large and often subject to licensing restrictions. The original study by Sirignano uses NASDAQ Level III data for 489 stocks, resulting in roughly 50 terabytes of raw data [7]. Such a dataset is far beyond the scope of this project, both in terms of access costs and computational requirements. In practice, obtaining a suitable limit order book dataset without institutional access is difficult. Commercial data providers such as LOBSTER offer high-quality reconstructed order book data, but access to larger datasets typically requires paid subscriptions or custom orders. This makes full-scale replication hard for a student project, where spending large amounts of money on data is not realistic. Several free or publicly available datasets are commonly used in academic work.

he FI-2010 dataset is a widely used benchmark for mid-price movement prediction from limit order book data [6]. It has been useful for comparing machine learning models, but it is relatively small and is mainly designed around a classification problem for future mid-price movement. LOBSTER also provides sample files, which are useful for understanding the data format and developing code, but these samples are too limited for a meaningful large-scale empirical comparison [3].

For this project, we use the WSELOB-2017 dataset [5]. This dataset contains order book data for the five largest companies listed on the Warsaw Stock Exchange over the year 2017. Compared with the LOBSTER sample files, WSELOB-2017 provides substantially more observations and multiple stocks, making it more suitable for testing whether the modelling ideas generalize across assets. At the same time, it is publicly available and stored in a practical HDF5 format, which makes it feasible to process within the computational constraints of this project. The choice of WSELOB-2017 also changes the interpretation of the replication.

We do not attempt to reproduce the numerical results of [7] exactly, since the market, data source and reconstruction procedure differ from the original NASDAQ setting. Instead, we use WSELOB-2017 to perform a reduced replication of the main modelling idea: estimating the conditional distribution of future best ask and best bid movements from the current state of the limit order book.

4.1.2. Structure of the WSELOB-2017 Dataset

The WSELOB-2017 dataset consists of limit order book data for five large stocks traded on the Warsaw Stock Exchange during 2017. Each stock is provided as an HDF5 file containing order book events for individual trading days. These events include information such as timestamps, prices, order side, aggregate volume and action type. From these event files, we reconstruct fixed-frequency order book snapshots that can be used as input for the prediction models.

The structure of WSELOB-2017 differs in an important way from the NASDAQ Level III data used in the original paper. The original dataset is closer to an event-by-event order-level record, containing detailed information about order submissions, cancellations and executions, together with the evolving state of the book. In contrast, the WSELOB data used in this project are provided as aggregate price-level updates. In our reconstruction, the variable `agg_volume` is interpreted as the total available volume at the affected price level after the event. Thus, the processed data describe how much liquidity is available at each observed price level, but they do not retain the full lifecycle of individual orders or detailed queue-position information.

This difference also affects the implementation. Since the data are naturally available at the aggregate price-level rather than individual-order level, our models are built from reconstructed book snapshots and price-level volume features. For the full local spatial neural network, this motivates the dense tick-grid representation: liquidity is placed on a regular grid indexed by tick distance from the current best ask or best bid, and local windows are extracted from this grid around candidate movement levels. This gives a practical approximation of the local spatial information used in the original paper, but it is less detailed than the original Level III setting. Consequently, the experiments should be interpreted as a reduced replication of the spatial modelling idea rather than as a direct reproduction of the original data environment.

In our experiments, we use the five stocks listed in Table 4.1. For each stock, we reconstruct snapshots at one-second intervals, using $L = 50$ levels on both sides of the book. The prediction horizon is fixed at $\Delta t = 30$ seconds, and movements are clipped to the range $[-10, 10]$ ticks. The table reports the number of reconstructed snapshots and the number of usable samples after removing observations for which no future snapshot is available at the prediction horizon.

Ticker	Company	Reconstructed snapshots	Usable samples
KGHM	KGHM Polska Miedź	$10{,}040{,}602$	$10{,}033{,}102$
PEKAO	Bank Pekao	$10{,}128{,}396$	$10{,}120{,}912$
PKNORLEN	PKN Orlen	$10{,}115{,}890$	$10{,}108{,}427$
PKOBP	PKO Bank Polski	$9{,}846{,}299$	$9{,}838{,}817$
PZU	Powszechny Zakład Ubezpieczeń	$9{,}989{,}786$	$9{,}982{,}286$

Table 4.1: Overview of the WSELOB-2017 stocks used in this project after reconstruction into one-second snapshots.

The dataset is large enough to train and compare several neural network models, with roughly ten million usable samples per stock. At the same time, it remains manageable on a local machine and does not require the storage and computational resources of the original NASDAQ Level III dataset used in [7].

4.1.3. Label Construction and Class Imbalance

For each reconstructed snapshot, we construct a label by comparing the best ask and best bid prices at the current time t with their values at the prediction horizon $t + \Delta t$. The resulting label is the clipped joint movement

$$Y_t = (\Delta A_t, \Delta B_t) \in \{-K, \dots, K\}^2.$$

In our experiments we use $K = 10$, so that the prediction problem has 441 possible joint movement classes. A central difficulty of this prediction problem is class imbalance. Over a fixed short horizon, the most common outcome is often that neither the best ask nor the best bid changes. This corresponds to the class $(0, 0)$. The extent of this imbalance differs substantially between stocks, as shown in Table 4.2. More active stocks have a larger fraction of nonzero movement observations, while less active stocks are dominated by the no-movement class. This imbalance is important for interpreting model performance. A model that predicts the no-movement class for almost every observation can obtain a high accuracy, especially for less active stocks such as PKOBP and PZU. For this reason, accuracy alone is not a reliable measure of predictive quality. In the evaluation, we therefore focus

Stock	Test samples	No-movement share	Movement share
KGHM	$1{,}504{,}966$	68.04%	31.96%
PEKAO	$1{,}518{,}137$	75.30%	24.70%
PKNORLEN	$1{,}516{,}265$	66.06%	33.94%
PKOBP	$1{,}475{,}823$	90.24%	9.76%
PZU	$1{,}497{,}344$	87.45%	12.55%

Table 4.2: Class imbalance in the chronological test split. Movement share denotes the proportion of observations for which $(\Delta A_t, \Delta B_t) \neq (0, 0)$.

primarily on negative log-likelihood, which measures the quality of the full predicted probability distribution. We also report movement-conditional negative log-likelihood, which evaluates performance only on observations where at least one side of the best quote changes.

4.2. Training Procedure

All models are trained and evaluated using a chronological split of the data. For each stock, the first 70% of observations are used for training, the next 15% for validation and the final 15% for testing. This split reflects the forecasting setting of the problem: models are fitted on earlier observations and evaluated on later, unseen observations.

For all conditional models, input features are standardized using only the training set. The resulting scaling parameters are then applied to the validation and test sets. This avoids information leakage from future observations. The empirical baseline does not use input features and is fitted only from the empirical class frequencies in the training set. The neural network models are trained with the Adam optimizer and weight decay. Dropout is used in the hidden layers as an additional form of regularization.

During training, model performance is monitored on the validation set after each epoch, and early stopping is used to select the final checkpoint. For the standard neural network and the reduced spatial neural network, we retain the checkpoint with the best validation performance according to the main training objective.

For the full local spatial neural network, we store two checkpoints: one selected using the full validation set and one selected using only validation observations with nonzero movement. This allows us to separately study overall calibration and performance on actual quote changes. After model selection, all final results are computed on the held-out test set. The test set is not used during training, validation or checkpoint selection.

4.3. Evaluation Metrics

The goal of this project is to estimate the full conditional distribution of future best ask and best bid movements. For this reason, the primary evaluation metric is negative log-likelihood. Let

$$\mathcal{D}_{\text{test}} = \{(X_i, Y_i)\}_{i=1}^{N_{\text{test}}}$$

denote the test set, and let $p_\theta(Y_i \mid X_i)$ be the probability assigned by a model to the observed label. The test negative log-likelihood is defined as

$$\mathrm{NLL} = -\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \log p_\theta(Y_i \mid X_i).$$

A lower value indicates that the model assigns higher probability to the realized future movements. This metric is more informative than simply measuring whether the most likely class is correct, because it evaluates the entire predicted probability distribution. Because the label distribution is highly imbalanced, we also report negative log-likelihood conditional on movement. Let

$$\mathcal{M} = \{i : Y_i \neq (0, 0)\}$$

be the set of test observations for which at least one side of the best quote changes. The movement-conditional negative log-likelihood is

$$\mathrm{MoveNLL} = -\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \log p_\theta(Y_i \mid X_i).$$

This metric measures how well the model performs on the more difficult and economically more interesting subset of observations where the best ask or best bid actually moves. It is especially useful because a model can obtain a good overall NLL by assigning high probability to the no-movement class, while still performing poorly on nonzero movements. For completeness, we also report classification accuracy. The predicted class is taken to be the class with highest predicted probability,

$$\hat{Y}_i = \arg\max_{y \in \mathcal{Y}} p_\theta(Y_i = y \mid X_i),$$

and the accuracy is

$$\mathrm{Accuracy} = \frac{1}{N_{\mathrm{test}}} \sum_{i=1}^{N_{\mathrm{test}}} \mathbf{1}\{\hat{Y}_i = Y_i\}.$$

However, accuracy must be interpreted with care. Since the no-movement class $(0, 0)$ is often dominant, a model that predicts no movement almost always can achieve high accuracy without providing a useful probabilistic model of price changes.

5 Implementation

This chapter briefly describes the software implementation used for the experiments. The code for the project is available on GitHub at `https://github.com/IdsRehorst/wi4224_deep_learning_for_limit_order_books.git`. The implementation consists of a preprocessing pipeline for reconstructing WSELOB-2017 limit order book snapshots, scripts for training the baseline and neural network models, and utilities for evaluating and comparing results.

5.1. Software Environment

The implementation was written in Python. Python was chosen because it provides easy to use libraries for numerical computing, data processing and neural network training. The main numerical operations are handled using NumPy, while pandas and PyTables are used to read and process the HDF5 files provided in the WSELOB-2017 dataset. Scikit-learn is used for preprocessing and for several evaluation utilities.

The neural network models are implemented in PyTorch. This includes the standard feedforward neural network, the reduced spatial neural network and the full local spatial neural network. PyTorch was chosen because it gives direct control over the model architectures and training loops, which was necessary for implementing the structured hazard-style output representations used in the spatial models.

The results of each experiment are saved as JSON files containing the model configuration, dataset metadata, training history and final evaluation metrics. This makes the experiments easier to compare and allows the tables and figures in this report to be generated directly from the saved result files.

5.2. Project Structure

The codebase is organized as a small experimental pipeline. The repository is available at `https://github.com/IdsRehorst/wi4224_deep_learning_for_limit_order_books.git`. The main folders are organized as follows:

`data/raw/`	raw input data files, including WSELOB-2017 HDF5 files,
`data/processed/`	processed datasets used by the training scripts,
`src/`	preprocessing and model training scripts,
`results/`	saved model outputs, metrics and checkpoints,
`figures/`	figures used in the report.

The raw WSELOB files are not included in the repository, since they are relatively large and should be downloaded separately from the original data source. The processed datasets and result files are generated by running the scripts in the `src/` folder.

The `src/` folder contains the main components of the pipeline. The preprocessing script converts raw WSELOB event data into fixed-frequency order book snapshots and prediction labels. Separate training scripts are then used for the empirical baseline, logistic regression, standard neural network, reduced spatial neural network and full local spatial neural network. Each training script writes its output to the `results/` directory, including the relevant hyperparameters and final evaluation metrics.

5.3. Preprocessing Pipeline

The preprocessing pipeline converts the raw WSELOB-2017 order book event files into datasets that can be used directly by the training scripts. This is done using the script `convert_wselob.py`. The raw files are provided in HDF5 format and contain order book events for each trading day. These events are first read day by day and used to reconstruct the state of the order book over time.

After reconstruction, the book is sampled at fixed one-second intervals. This gives a regular sequence of order book snapshots, which makes the prediction problem easier to define and compare across stocks. For each snapshot, we store the first $L = 50$ levels on both the ask and bid side. From these levels, we construct the flat feature representation used by the empirical baseline, logistic regression, the standard neural network and the reduced spatial neural network. These features include transformed ask and bid volumes, level-wise imbalance features, relative price offsets and the spread.

The next step is label construction. For each snapshot at time t, the best ask and best bid are compared with their values at time $t + \Delta t$, where $\Delta t = 30$ seconds in the final experiments. The resulting ask and bid movements are measured in tick units and clipped to the range $[-10, 10]$. Observations for which no future snapshot is available at the required horizon are removed.

For the full local spatial neural network, an additional dense representation is saved. Instead of only storing the observed book levels, the ask and bid volumes are placed on a dense tick grid relative to the current best ask and best bid. Missing price levels are assigned zero volume. This representation is needed because the full spatial model extracts local windows around candidate price movement levels.

The final output of the preprocessing step is a processed dataset folder for each stock. Each folder contains the feature arrays, labels, metadata and, when required, dense spatial arrays. The metadata file records the main preprocessing choices, including the number of levels, sampling interval, prediction horizon, clipping range and tick size. This makes it possible to verify that all models are trained on the same prediction problem.

5.4. Training Scripts

After preprocessing, each model is trained using a separate script in the `src/` folder. This keeps the implementation modular: all models use the same processed datasets, but each script contains the training logic specific to one model class.

The empirical baseline is implemented in `train_empirical.py`. This script estimates the unconditional class distribution from the training labels and evaluates the resulting probabilities on the validation and test sets. Since this model does not use the order book features, it serves as a simple reference point for the other models.

The logistic regression baseline is implemented in `train_logistic.py`. Because the full-year WSELOB datasets contain around ten million samples per stock, the logistic model is trained using mini-batch optimization rather than fitting all data in memory at once. This makes the baseline feasible to run on the full datasets while keeping the probabilistic softmax formulation.

The standard neural network is trained using `train_mlp.py`. This script fits a feedforward neural network with two hidden layers and a softmax output over the 441 joint movement classes. The reduced spatial neural network is trained using `train_spatial.py`. This script implements the factorized hazard-style output structure described in Section 3.3.2.

Finally, the full local spatial neural network is trained using `train_full_spatial.py`. This script uses the dense spatial arrays produced during preprocessing and extracts local windows around candidate tick levels. Because this model is more computationally expensive and more sensitive to class imbalance, it saves two checkpoints: one selected by validation NLL and one selected by validation movement NLL.

All training scripts save their results as JSON files in the `results/` directory. These files include the model configuration, training history, dataset metadata and final train, validation and test metrics. The neural network scripts also save trained model checkpoints. This makes it possible to inspect the experiments afterwards and to generate the result tables and figures in the report directly from the saved outputs.

5.5. Reproducibility

To improve reproducibility, fixed random seeds are used throughout the training scripts. This reduces variation due to random initialization and batch ordering, making the experiments easier to rerun and compare.

6

Results and Discussion

This chapter presents the empirical results of the model comparison. We first briefly recall the role of class imbalance in interpreting the results. We then compare the empirical baseline, logistic regression, standard neural network and reduced spatial neural network across all five stocks. Finally, we discuss the full local spatial network extension and interpret the main findings in relation to the original paper.

6.1. Label Distributions and Prediction Difficulty

As discussed in Section 4.1.3, the fixed-horizon prediction problem is highly imbalanced. For all stocks, the no-movement class $(0, 0)$ is the most common outcome, although the movement share differs substantially between assets. This means that the difficulty of the prediction task is not the same for every stock: for less active stocks, a large fraction of observations can be predicted correctly by simply assigning most probability to no movement. This imbalance is important when interpreting the results. In particular, classification accuracy can be misleading, since a model that predicts $(0, 0)$ almost always may still obtain a high accuracy. For this reason, as is mentioned in Section 4.3, the main comparison below focuses on negative log-likelihood and movement-conditional negative log-likelihood. These metrics better reflect the goal of estimating the full conditional distribution of future best ask and best bid movements.

6.2. Main Multi-Stock Model Comparison

We first compare the empirical baseline, logistic regression, the standard neural network and the reduced spatial neural network across all five WSELOB-2017 stocks. The full local spatial network is discussed separately in Section 6.3, since it is an additional extension with two checkpoint selection criteria.

Table 6.1 reports the test negative log-likelihood for the main models. The empirical baseline provides the unconditional reference point, while logistic regression, the standard neural network and the reduced spatial neural network use the current order book state as input.

Stock	Empirical	Logistic	Standard NN	Reduced spatial NN
KGHM	1.6584	1.4711	**1.3707**	1.3798
PEKAO	1.4069	1.2788	**1.2060**	1.2095
PKNORLEN	1.8856	1.7369	1.5931	**1.5922**
PKOBP	0.5453	0.5151	**0.4755**	0.4824
PZU	0.6838	0.6665	0.6263	**0.6044**

Table 6.1: Test negative log-likelihood for the main model comparison. Lower values are better.

The results show a clear improvement over the empirical baseline for every stock. Logistic regression already reduces the negative log-likelihood, which indicates that the current order book state contains useful predictive information. The neural network models improve further in all cases. The standard neural network achieves the best overall NLL for KGHM, PEKAO and PKOBP, while the reduced spatial neural network performs best for PKNORLEN and PZU. Overall, the standard neural network and reduced spatial neural network are very close, suggesting that both nonlinear modelling and structured output modelling are useful for this task.

Table 6.2 reports the movement-conditional negative log-likelihood. This metric evaluates the models only on observations where at least one side of the best quote changes.

Stock	Empirical	Logistic	Standard NN	Reduced spatial NN
KGHM	4.2929	3.6604	3.4976	**3.4254**
PEKAO	4.8358	4.2660	3.9864	**3.9764**
PKNORLEN	4.9786	4.2021	3.8439	**3.8364**
PKOBP	4.5559	4.2717	3.8940	**3.6885**
PZU	4.6169	4.2362	**3.8172**	3.8255

Table 6.2: Test movement-conditional negative log-likelihood for the main model comparison. Lower values are better.

The movement-conditional results are particularly important because the prediction problem is dominated by the no-movement class. Here, the reduced spatial neural network performs especially well. It achieves the best movement NLL for four out of the five stocks, with PZU being the only case where the standard neural network is slightly better. This suggests that the structured hazard-style output representation is useful for assigning probability mass to nonzero quote movements.

6.3. Full Local Spatial Network Extension

We also evaluate the full local spatial neural network introduced in Section 3.3.3. This model is closest to the spatial architecture proposed in [7], since it computes continuation probabilities from local liquidity windows around candidate price levels. In the original paper, the spatial neural network is found to outperform the standard neural network by exploiting the local structure of the limit order book. In our setting, we investigate whether the same idea remains effective on the reconstructed WSELOB-2017 data.

Because the full local spatial network is more complex and the dataset is highly imbalanced, we report two checkpoint selection criteria. The first checkpoint is selected by the best overall validation NLL. The second checkpoint is selected by the best validation movement NLL, using only observations for which $(\Delta A_t, \Delta B_t) \neq (0, 0)$.

Stock	Checkpoint selection	Test NLL	Test MoveNLL
KGHM	Validation NLL	1.3762	3.5728
KGHM	Validation MoveNLL	1.4444	**3.4184**
PEKAO	Validation NLL	**1.1982**	4.0087
PEKAO	Validation MoveNLL	1.2274	**3.8008**
PKNORLEN	Validation NLL	1.6316	4.2227
PKNORLEN	Validation MoveNLL	**1.6244**	**4.1684**
PKOBP	Validation NLL	12.3108	4.8688
PKOBP	Validation MoveNLL	**8.8620**	**3.9928**
PZU	Validation NLL	**1.1621**	4.2178
PZU	Validation MoveNLL	3.1346	**3.8274**

Table 6.3: Test performance of the full local spatial neural network under two checkpoint selection criteria. Lower values are better.

The results do not reproduce the strong dominance of the spatial neural network reported in [7]. Instead, the effect of the full local spatial model is mixed. For several stocks, selecting the checkpoint by movement NLL improves performance on nonzero quote movements, but this often comes at the cost of worse overall NLL. This trade-off is especially relevant because WSELOB-2017 contains several less active limit order books, where the no-movement class dominates the fixed-horizon prediction problem.

The difference is most visible for stocks such as PKOBP and PZU. These stocks have substantially lower movement shares than the more active stocks, so assigning probability mass to movement classes can strongly hurt the overall likelihood. This suggests that the usefulness of a full local spatial architecture depends not only on the model design, but also on the activity level of the underlying limit order book and on the evaluation criterion used.

6.4. Qualitative Analysis of Predicted Distributions

To complement the quantitative evaluation, it is useful to inspect the predicted class probabilities directly. Figure 6.1 shows the average predicted joint distribution over clipped quote movements for KGHM, computed over those test observations for which the realized movement is nonzero. Each heatmap therefore summarizes how a model distributes probability mass over the grid

$$\{-K, \dots, K\}^2,$$

with the horizontal axis corresponding to the bid movement ΔB_t and the vertical axis to the ask movement ΔA_t.

Several features of the figure are informative. First, all three models place substantial probability mass near the center of the grid, including around $(0, 0)$. This is not surprising, since the no-movement class remains the dominant class in the data overall, and even when conditioning on movement observations the models retain part of this prior tendency. At the same time, the heatmaps reveal clear qualitative differences in how the remaining mass is allocated.

The standard neural network produces the most diffuse distribution. Although it recognizes that movements with ΔA_t and ΔB_t of similar sign are relatively likely, it still spreads noticeable probability mass over a broad region of the grid. In particular, the distribution is relatively smooth and does not strongly suppress less plausible parts of the output space. This is consistent with the architecture described in Section 3.3.1: the model is nonlinear, but it still treats the 441 classes as separate categories and does not explicitly encode spatial relations between neighboring movement classes.

The reduced spatial neural network produces a more structured distribution. Probability mass is concentrated more strongly along the economically plausible region of the grid, especially around the diagonal corresponding to joint upward and downward quote movements. At the same time, the model assigns much less mass to the off-diagonal quadrants where the ask and bid move in opposite directions by large amounts. This is a direct consequence of the hazard-style construction from Section 3.3.2, which introduces an ordered representation of movement size and makes nearby classes statistically related.

The full local spatial neural network sharpens this effect further. Its heatmap is the most structured of the three: the mass is concentrated in a narrower region, and implausible areas of the grid are suppressed more aggressively. This is consistent with the local spatial construction from Section 3.3.3. Since continuation probabilities are computed from local liquidity windows around candidate price levels, the model is encouraged to assign probability in a way that respects the local geometry of the order book. In this sense, Figure 6.1 provides visual evidence that the full local spatial model is not merely a different classifier, but that it produces a more coherent spatial distribution over quote movements.

These heatmaps also help explain why movement-conditional negative log-likelihood is an informative metric in this study. The main advantage of the spatial models is not necessarily that they dramatically change the most likely class, but rather that they redistribute probability mass more sensibly over nearby nonzero movement classes. This is precisely what Figure 6.1 illustrates: compared with the standard neural network, the spatial models produce a more concentrated and more interpretable distribution on the movement observations.

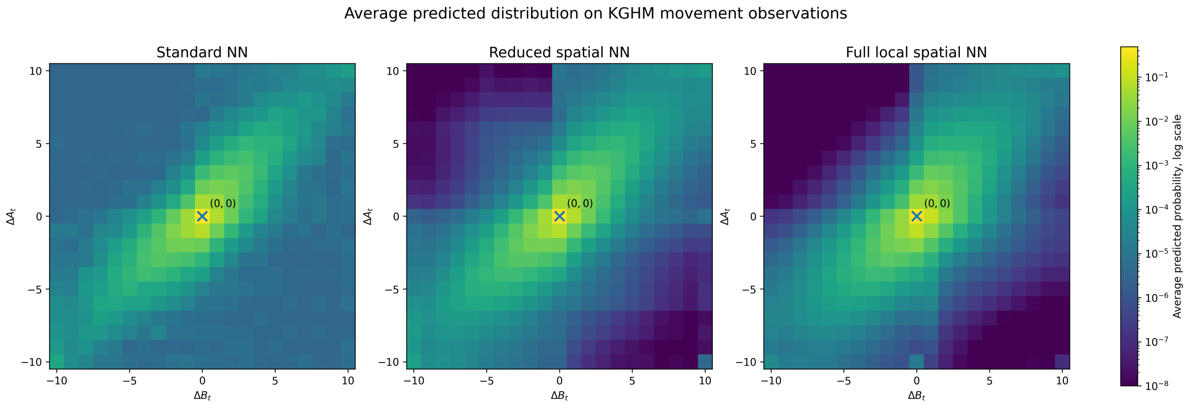

Figure 6.1: Average predicted joint distribution of $(\Delta A_t, \Delta B_t)$ on KGHM test observations with nonzero realized movement. The standard neural network produces a relatively diffuse distribution, whereas the reduced spatial and full local spatial models concentrate probability mass more strongly along plausible regions of the output grid. The marker indicates the no-movement class $(0, 0)$. Colors are shown on a logarithmic scale.

6.5. Interpretation and Limitations

The results show that the current limit order book state contains useful information for predicting future best ask and best bid movements. For every stock, logistic regression improves upon the empirical baseline, which indicates that the input features have predictive value beyond the unconditional class distribution. The neural network models improve further, showing that nonlinear relationships between book depth, spread, imbalance and relative price levels are relevant for the prediction task.

Figure 6.2 provides a compact visual summary of the movement-conditional negative log-likelihood results. The figure makes clear that the neural network models consistently outperform the empirical and logistic baselines on nonzero quote movements. It also shows that the reduced spatial neural network is particularly competitive on this metric, achieving the best movement NLL for most stocks in the main comparison. The full local spatial network improves movement NLL for some stocks, especially KGHM and PEKAO, but its performance is less uniform across the full set of assets.

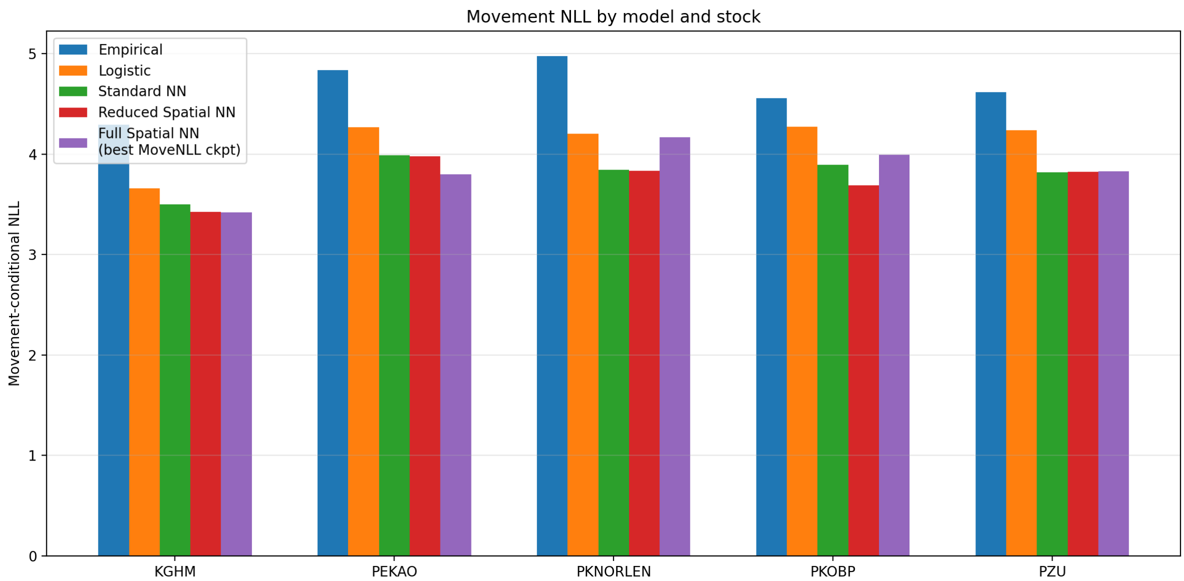

Figure 6.2: Movement-conditional negative log-likelihood across stocks and models. Lower values are better. The figure highlights that the spatial models are especially relevant when evaluation is restricted to observations with nonzero quote movements.

The standard neural network and the reduced spatial neural network are the most reliable models in the main comparison. The standard neural network performs well in terms of overall negative log-likelihood, which suggests good calibration across the full test set. The reduced spatial neural network is especially strong on movement-conditional negative log-likelihood. This supports the idea that adding structure to the output space can help the model assign probability to nonzero quote movements.

At the same time, the results do not fully reproduce the main finding of [7], where the spatial neural network strongly outperforms the standard neural network. In our experiments, the spatial models are competitive, but they do not uniformly dominate the standard neural network. This difference is likely related to the reduced scale of our replication, the different market setting and the lower activity of some WSELOB limit order books. In less active stocks, the no-movement class dominates the fixed-horizon prediction problem, making it difficult for more complex spatial models to improve overall likelihood.

The full local spatial neural network highlights this trade-off most clearly. When selected by overall validation NLL, it tends to behave conservatively and assigns most probability to no movement. When selected by movement-conditional validation NLL, it improves performance on nonzero movements for several stocks, but often worsens the overall NLL. The strongest movement-NLL improvements occur for KGHM and PEKAO, which are among the more active order books in our dataset. By contrast, for less active stocks such as PKOBP and PZU, where the no-movement class dominates more strongly, the full spatial model becomes less stable and does not clearly improve the main comparison. PKNORLEN is an exception to a simple activity-based explanation, but the overall pattern suggests that the usefulness of the full local spatial architecture depends partly on how often informative movement events occur.

These findings also explain why accuracy is not a suitable primary metric for this problem. Several models predict the no-movement class for the vast majority of observations, which can lead to high accuracy on inactive stocks. However, this does not imply that the model captures the distribution of price changes well. Negative log-likelihood and movement-conditional negative log-likelihood provide a more informative evaluation because they measure the probabilities assigned to the realized outcomes.

There are several limitations to the experiment. First, the scale of the data is much smaller than in the original study. Sirignano [7] uses NASDAQ Level III data for hundreds of stocks and a substantially larger raw dataset, whereas our experiments are based on five reconstructed WSELOB-2017 stocks. Although each WSELOB stock still contains millions of observations, the cross-sectional diversity and total amount of training data are much smaller. This is especially relevant for the spatial models, since they introduce stronger structure and more specialized components. With less data, it is harder to train these models reliably and to expect the same level of generalization as in the original large-scale setting.

Second, our output space is a finite approximation of the original prediction problem. The original spatial neural network is designed to model a distribution over future price locations without restricting the output to a small fixed grid. In contrast, we clip ask and bid movements to $\{-K, \dots, K\}^2$, with $K = 10$, so the model predicts over 441 joint movement classes. This makes the problem computationally feasible and allows direct comparison between empirical, logistic, standard neural network and spatial models. However, it also means that large movements are not modelled explicitly because they are absorbed into boundary classes. As a result, our models do not estimate a full distribution on $\mathbb{R}^2$, but only a truncated and discretized approximation of the joint movement distribution.

Third, the WSELOB-2017 data are reconstructed from aggregate order book events and do not correspond exactly to the NASDAQ Level III data used in the original paper. The processed WSELOB representation describes aggregate liquidity available at price levels, but it does not retain the same individual order-level and queue-position information. This affects the amount of local information available to the full spatial model.

Fourth, we use a fixed prediction horizon of 30 seconds for all stocks. This keeps the experimental setup simple and comparable across assets, and follows the spirit of the fixed-horizon formulation in the original paper. However, it is not equally suitable for every stock. The WSELOB stocks differ substantially in activity level, so the same clock-time horizon produces many more no-movement observations for less active stocks. A more refined setup would use stock-specific horizons or event-time horizons, for example predicting after a fixed number of quote updates or order book events.

Finally, the full local spatial architecture is still an approximation of the original spatial neural network. Our implementation uses fixed local windows on a dense tick grid and a clipped finite output space. This captures the main spatial intuition that continuation probabilities should depend on nearby liquidity, but it is not a complete reproduction of the original architecture. For these reasons, the results should be interpreted as evidence about how much of the spatial modelling idea carries over to a smaller public-data setting, rather than as a direct empirical replication of the original study.

A natural question is why the prediction horizon was not chosen to be longer. A longer horizon would indeed reduce the no-movement imbalance, since the best ask and best bid would have more time to change. However, this would also change the nature of the prediction problem. At a short horizon, the current limit order book state X_t is still expected to contain information about the future quote movement, because the prediction is closely tied to the liquidity currently available near the best bid and ask. At a longer horizon, many new orders, cancellations and trades may occur between t and $t + \Delta t$. The prediction problem then becomes less local because it is no longer only about how the current book state is consumed, but also about future order flow that is not observed at time t. In addition, longer horizons would lead to larger price movements more often, so the clipped grid $\{-K, \ldots, K\}^2$ would become more restrictive and boundary classes would absorb more observations.

We use the same horizon for each stock mainly to keep the experimental setup comparable across assets and to stay close to the fixed-horizon formulation used in the original paper. This choice makes the results easier to interpret, since all stocks are evaluated on the same prediction task. At the same time, it is a clear limitation. The WSELOB stocks have different activity levels, so a horizon of 30 seconds does not represent the same amount of market activity for every asset. For active stocks, 30 seconds may contain many informative quote updates, while for less active stocks the same horizon often results in no movement. This partly explains why the no-movement class dominates much more strongly for some stocks than for others. A more refined design would use stock-specific horizons or event-time horizons, for example predicting after a fixed number of order book events or quote updates. Such a design would make the prediction problem more comparable in terms of market activity, but it would also move further away from the simple fixed-horizon setup used in this reduced replication.

Overall, the reduced replication provides partial support for the modelling ideas in [7]. Neural networks clearly improve upon simple baselines, and spatial structure is useful for modelling movement observations. However, on the WSELOB-2017 data, the benefits of spatial modelling are more nuanced than in the original study: the reduced spatial network is a stable and useful extension, while the full local spatial network mainly helps when the evaluation focuses specifically on nonzero quote movements.

7 Conclusion

This project studied whether neural network models can be used to predict short-term limit order book movements. Following the formulation of [7], we focused on the joint distribution of future best ask and best bid movements, rather than only predicting the direction of the mid-price. Due to data and computational constraints, the goal was not to reproduce the original study exactly, but to perform a reduced replication on the publicly available WSELOB-2017 dataset.

The experiments show that the current state of the limit order book contains useful predictive information. For every stock, logistic regression improves upon the naive empirical baseline, indicating that the constructed order book features have predictive value beyond the unconditional label distribution. The neural network models improve further, showing that nonlinear relationships between book depth, spread, imbalance and relative price levels are relevant for the prediction task.

The standard neural network and the reduced spatial neural network are the strongest models in the main comparison. The standard neural network performs well in terms of overall negative log-likelihood, while the reduced spatial neural network is especially competitive on movement-conditional negative log-likelihood. This suggests that adding structure to the output space can help when predicting nonzero quote movements. The full local spatial neural network provides additional support for this idea, but its benefits are more mixed. When selected specifically for movement-conditional validation performance, it improves movement likelihood for some stocks, especially the more active ones, but this often comes at the cost of worse overall calibration.

Compared with the original paper, our results are therefore more nuanced. We do not find that the spatial neural network uniformly dominates the standard neural network. Instead, spatial structure appears most useful when the evaluation focuses on observations where the best ask or best bid actually changes. This difference is likely related to the smaller scale of the experiment, the different market setting and the relatively inactive nature of some WSELOB limit order books.

The main limitation of this project is that the WSELOB-2017 data are reconstructed from aggregate order book events and do not correspond exactly to the NASDAQ Level III data used in the original study. In addition, we use a fixed prediction horizon of 30 seconds for all stocks, even though the average time between quote changes differs across assets. A more refined setup could use stock-specific or event-time horizons, which may make the prediction problem more comparable across stocks.

Overall, this reduced replication provides partial support for the modelling ideas in [7]. Neural networks clearly improve upon simple baselines, and spatial structure can improve the modelling of actual quote movements. At the same time, the results show that the usefulness of spatial neural networks depends strongly on the dataset, the activity level of the order book and the evaluation criterion. Future work could investigate more faithful implementations of the original spatial architecture, event-time prediction horizons, online model updating and larger or more active limit order book datasets.

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